

Skeleton2Stroke: Interactive Stroke Correspondence Editing with Pose Features

Ryoma Miyauchi
Yichen Peng
ryoma.riruru@gmail.com
yichen.peng@jaist.ac.jp
Japan Advanced Institute of Science
and Technology
Japan

Tsukasa Fukusato
tsukasafukusato@is.s.u-tokyo.ac.jp
The University of Tokyo
Japan

Haoran Xie
xie@jaist.ac.jp
Japan Advanced Institute of Science
and Technology
Japan

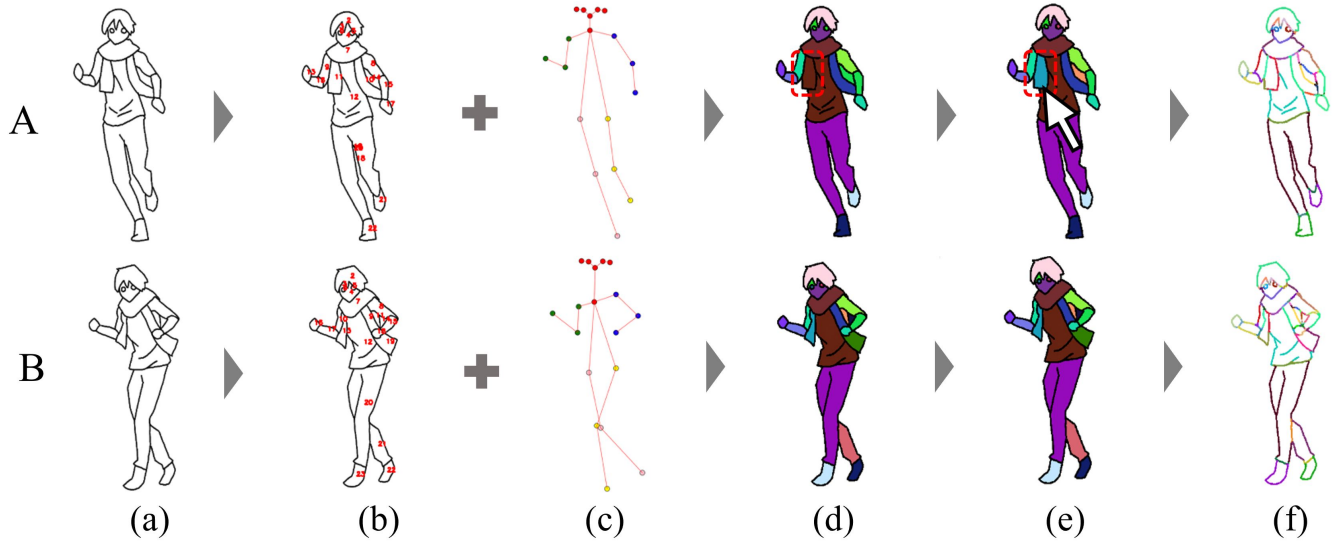


Figure 1: System overview: (a) Input source images, (b) closed region with shape features, (c) skeleton estimation with pose features, (d) estimation of the closed-areas correspondence, (e) user editing, and (f) the final results of stroke correspondence.

ABSTRACT

Inbetweening is an important technique for computer animations where the stroke correspondence of hand-drawn illustrations plays a significant role. Previous works typically require image vectorization and enormous computation cost to achieve this goal. In this paper, we propose an interactive method to construct stroke correspondences in character illustrations. First, we utilize a deep learning-based skeleton estimation to improve the accuracy of closed-area correspondences, which are obtained using greedy algorithm. Second, we construct stroke correspondences based on the estimated closed-area correspondences. The proposed user interface

is verified by our experiment to ensure that the users can achieve high accuracy with low correction in stroke correspondence.

CCS CONCEPTS

• **Computing methodologies** → **Graphics systems and interfaces**; Animation.

KEYWORDS

Stroke correspondence, closed-area correspondence, inbetweening, pose estimation, user interface

ACM Reference Format:

Ryoma Miyauchi, Yichen Peng, Tsukasa Fukusato, and Haoran Xie. 2021. Skeleton2Stroke: Interactive Stroke Correspondence Editing with Pose Features. In *SIGGRAPH Asia 2021 Technical Communications (SA '21 Technical Communications)*, December 14–17, 2021, Tokyo, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3478512.3488612>

1 INTRODUCTION

The traditional production pipeline of hand-drawn animations (e.g., Japanese anime) is usually divided into two processes: drawing of the key frames, including the feature points of the characters'

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
SA '21 Technical Communications, December 14–17, 2021, Tokyo, Japan

© 2021 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-9073-6/21/12...\$15.00
<https://doi.org/10.1145/3478512.3488612>

movements, and drawing inbetween frames as the interpolation of key frames. However, inbetweening production takes an enormous amount of labor and cost for animators. To solve this issue, the automatic inbetweening approaches have been proposed to reduce the time effort [Fukusato and Morishima 2016; Yang 2017]. Conventional approaches require animators to vectorize hand-drawn frames and construct stroke-by-stroke correspondences with commercial software whose operations are tedious and inefficient. We especially focus on character drawings as source images due to the natural and representative features in animation productions.

In this work, we propose Skeleton2Stroke, an interactive method for directly constructing stroke correspondences for hand-drawn character source illustration (raster images) as shown in Figure 1. First, we estimate the closed-area correspondences from the source images, considering their shape and pose features. In addition, we incorporate the user's editing process into an optimization loop to achieve highly accurate estimation results. In our experiment, we compare the baseline approaches that consider only shape features to the proposed method in terms of the number of user corrections, and the percentage of accuracy of closed-area correspondences

2 RELATED WORKS

Generating inbetween frames requires stroke correspondences between the two input key frame drawings. However, manual correspondence setting is time consuming and labor intensive for users, since character drawings consist of hundreds of strokes. Hence, several studies have considered the automatic construction of stroke correspondences between images by imposing several constraints on stroke drawing orders and directions. For example, Yang et al. [2018] employed a greedy algorithm that used stroke characteristics, such as stroke connection, shapes, and locations. However, these methods require re-drawing images with vector lines and annotating strokes with spatial and directional information.

Meanwhile, the drawing processing plays an important role in anime production, which includes automatic colorization [Zhang et al. 2018], physical effect [Hu et al. 2019], and photo-realistic image synthesis [Huang et al. 2021] for line drawings. In this work, we focus on *closed areas* in character drawings and extend them to construct stroke correspondences for inbetweening production. In addition, we combine character pose features to improve the accuracy of closed-area correspondence construction.

3 SKELETON2STROKE

Figure 1 provides a system overview for constructing stroke correspondences between two character illustrations. Note that, as pre-processing, we employ a Gaussian filter and a fine-line filter [Zhang and Suen 1984] to remove noises and compose strokes with near-constant width. First, we label the closed areas of each image and compute a score for each closed area based on front-back relationship (depth), neighbor relationship, and posture information. Based on the scores, the system estimates closed-area correspondences using a greedy algorithm. Using the estimated closed-area correspondence, stroke correspondences are estimated automatically.

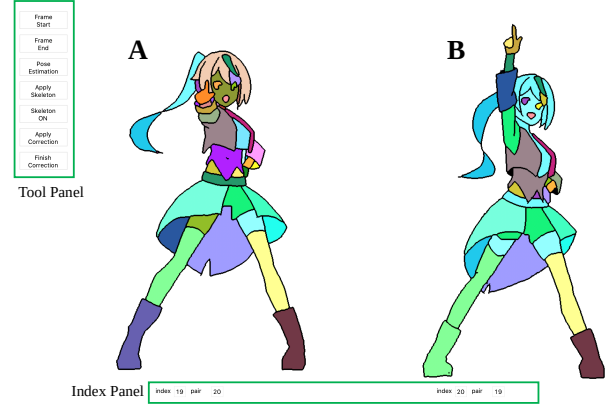


Figure 2: The editing interface of Skeleton2Stroke system.

3.1 Editing Interface

Figure 2 illustrates the editing interface of the proposed stroke correspondence construction. The proposed interface has tool and index panels. In the tool panel, we provide the following functions: loading two images (A and B), pose estimation, closed-area correspondence with the pose features, visualization of character postures, and editing closed-area correspondences to achieve stroke correspondences. The index panel displays the input images (left: image A, right: image B), and the currently selected closed-area identification number.

3.2 Closed-area Correspondence

To estimate correspondences between closed-areas in two images, we first assign $a_i^A, i \in \{0, \dots, N_a^A\}$ to be the closed area in image A and $a_j^B, j \in \{0, \dots, N_a^B\}$ to be the closed areas in image B. As with Yang et al. [2018], we use the greedy algorithm to produce optimal correspondence by continuously finding local optima. We compute a score $E(i, j)$ between a_i^A and a_j^B as follows:

$$E(i, j) = E_s(i, j) + E_p(i, j) \quad (1)$$

where E_s is a score computed with shape features (i.e., shape, depth, and region connections of closed regions). Please refer to [Miyauchi et al. 2021] for more details. E_p is computed with pose features.

3.2.1 Character Pose Features. We extract the 2D skeleton poses (i.e., joints and bones) of hand-drawn characters from images A and B and update the scores under the skeletons to improve the estimation accuracy of closed-area correspondence. We utilize a deep learning-based pose estimation to obtain human poses from a character drawing. Due to the monochromatic nature of line drawings, we decided to adopt AlphaPose [Fang et al. 2017] for estimating posture from character drawings, as shown in Figure 3. AlphaPose ensures sufficient detection accuracy of the existing Single Person Pose Estimation (SPPE) model by separating the character's detected position and posture from images. However, the estimated postures may lack accuracy when the characters is wearing loose clothes that hide their figure. To overcome this issue, we make a simple function to manually edit the estimated bone structures using a drag-and-drop operation (or add/delete bones).

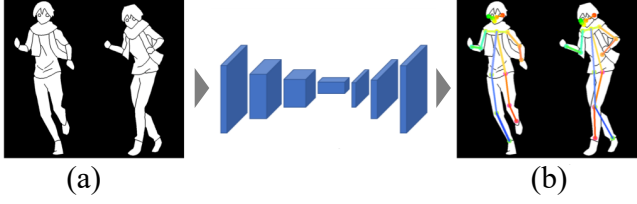


Figure 3: Pose estimation using AlphaPose [Fang et al. 2017]. (a) Input character images and (b) the estimated postures.

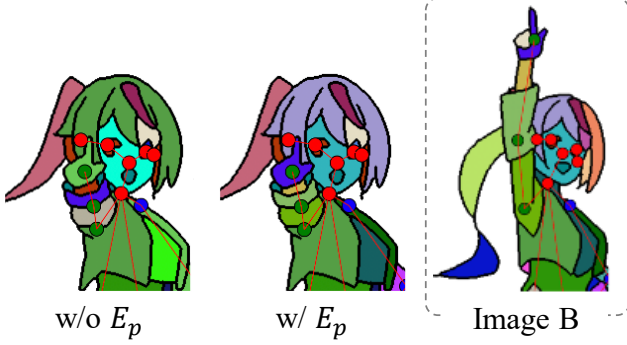


Figure 4: An example of closed-area correspondence considering pose features. Note that correspondences neighboring the right arm are updated by referring to two postures.

We sample M points (in this paper, we set $M = 50$) uniformly distributed on each bone and assign a closed-area ID to each point. Let p_m^l and q_m^l be a closed-area ID assigned to m -th point of l -th bone, respectively, in images A and B. Next, we compute a pose feature $E_p(i, j)$ based on the sampling points p_m^l and q_m^l as follows:

$$E_p(i, j) = \sum_{l \in \text{bone}} \sum_{m=1}^M \sum_{n=m-r}^{m+r} \|\phi_l(m, n) + \phi_l(n, m)\| \quad (2)$$

$$\phi_l(x, y) = \begin{cases} 1 & \text{if } p_x^l = i \wedge q_y^l = j \\ 0 & \text{else} \end{cases}$$

where r is a constant value (in this paper, we empirically set $r = 2$). Figure 4 shows an example of estimated closed-area correspondences with the pose features E_p .

3.2.2 User Editing. In our editing interface (Figure 2), users can directly modify the correspondences of closed areas. Users may specify partial correspondences and the system will then assign a large score to the selected regions and re-optimizes the correspondences (see Equation 1).

3.3 Stroke Correspondence

The second step is to estimate stroke correspondences using the estimated closed-area correspondences. In this paper, we regard the boundary lines of closed areas in images A and B as strokes and focus on connection relationships between closed areas. Let s_i^A , $i \in \{0, \dots, N_s^A\}$ be strokes in the image A and s_j^B , $j \in \{0, \dots, N_s^B\}$ be strokes in the image B. For example, when two closed areas in

the image A (a_1^A and a_2^A) are separated by one stroke s_0^A and two closed areas in the image B (a_1^B and a_2^B) are separated by one stroke s_0^B , it is thought that the stroke s_0^A corresponds to the stroke s_0^B . By repeatedly finding such correspondences, the proposed approach can easily construct stroke correspondences between two images without manual labor. Note that we employ Zhang's method [1984] in advance to segment closed areas in images A and B to strokes.

The proposed method allows animators to skip the manual re-drawing process (i.e., tracing over the hand-drawn illustrations), enabling users to focus on designing the movements of each stroke.

4 RESULTS

In order to verify the system effectiveness, we compared the accuracy of closed-area correspondences produced using three methods: (a) the shape features only ($= E_s$ only), (b) the pose features only ($= E_p$), and (c) our method ($= E_s + E_p$). Specifically, we applied these methods to four cases of hand-drawn character images available on video sites: (1) with little motion, (2) with large motion, (3) with little motion and body hidden by clothes, and (4) with large motion and body largely hidden by clothes. We then measured the number of user corrections needed until the correct closed-area correspondences were obtained. In addition, we computed the matching accuracy of stroke correspondences based on the correct closed-area correspondences.

Figure 5 and Table 1 show the estimated results. According to the results, we confirmed that our method ($= E_s + E_p$) achieves plausible correspondence estimation. However, the number of user corrections needed in cases 3 and 4 was higher than others. In addition, while there was almost no difference in the accuracy of stroke correspondence ($\approx 80\%$), the matching accuracy of stroke correspondence for case 4 was lower. One possible reason for this result is that the system found it difficult to handle self-occluded parts in loose clothes (e.g., changes in the number of closed areas/strokes).

4.1 Discussion

Although we utilized AlphaPose [Fang et al. 2017] to extract pose features from hand-drawn character images, it remains difficult to fully automate the estimation in cases where characters are wearing loose clothes that hide their figures (e.g., scarves and skirts in cases 3&4). Our prototype system enables users to manually edit features one-by-one, but the process is tedious and requires much time-consuming manual work. To address this issue, it might be interesting to extend interactive support for closed areas with scribble-based tools.

The proposed method may be challenged by the difficulty of estimating internal strokes in closed areas (e.g., cloth wrinkles) and self-occluded parts (e.g., change in the number of strokes/areas due to occlusion). We would like to combine our approach with the direct estimation of stroke correspondences [Yang et al. 2018].

5 CONCLUSION

In this paper, we proposed Skeleton2Stroke, an editing user interface that allows interactive construction of stroke correspondences between two hand-drawn character illustrations (raster images) based on closed-area correspondences estimated by shape and pose features. In addition, our prototype system allows users to directly edit

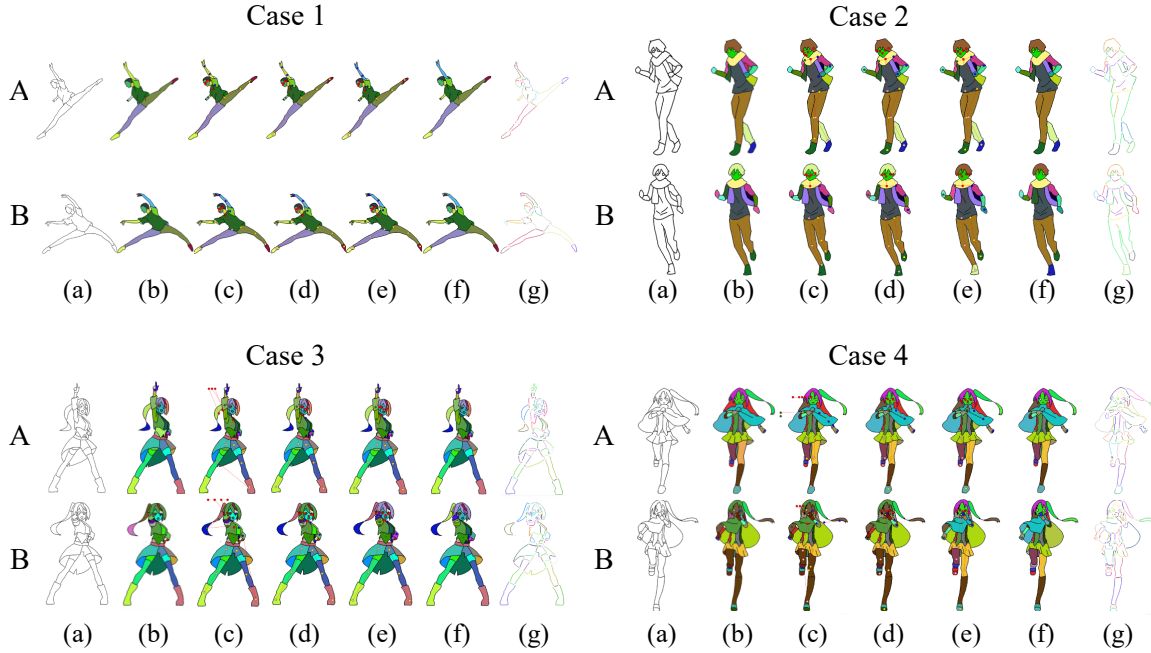


Figure 5: Experimental results. (a) Input two images A and B, (b) closed-area correspondences based on shape features (without pose features), (c) skeleton estimation results [Fang et al. 2017], (d) with user correction, (e) closed-area correspondences determined by combining the pose features, (f) with user correction, and (g) stroke correspondence using the closed-area correspondences. Note that the pairs of regions/strokes are in the same color.

Table 1: Experiment results with image statistics and correspondence accuracy.

Case ID	image statistics				#user correction (closed-area)			#accuracy of stroke correspondence [%]
	N_a^A	N_a^B	N_s^A	N_s^B	E_s only	E_p only	$E_s + E_p$	
1	19	19	66	61	5	6	0	86.4
2	23	20	67	66	7	8	2	89.4
3	46	48	132	134	19	32	7	88.2
4	49	53	149	171	24	35	17	78.9

closed-area correspondences and indirectly improve the accuracy of stroke correspondences. By comparing with baseline approaches that consider only shape features, we confirmed that our proposed system is effective for constructing stroke correspondences.

ACKNOWLEDGMENTS

This work was supported by Grant from Tateishi Science and Technology Foundation, JSPS KAKENHI JP20K19845 and JP19K20316.

REFERENCES

- Hao-Shu Fang, Shuqin Xie, Yu-Wing Tai, and Cewu Lu. 2017. RMPE: Regional multi-person pose estimation. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. IEEE, Venice, Italy, 2334–2343. <https://doi.org/10.1109/ICCV.2017.256>
- Tsukasa Fukusato and Shigeo Morishima. 2016. Active comicing for freehand drawing animation. In *Mathematical Progress in Expressive Image Synthesis III*. Springer, Singapore, 45–56. https://doi.org/10.1007/978-981-10-1076-7_6
- Zhongyuan Hu, Haoran Xie, Tsukasa Fukusato, Takahiro Sato, and Takeo Igarashi. 2019. Sketch2VF: Sketch-based flow design with conditional generative adversarial network. *Computer Animation and Virtual Worlds* 30, 3-4 (2019), e1889. <https://doi.org/10.1002/cav.1889> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/cav.1889> e1889 cav.1889.
- Zhengyu Huang, Yichen Peng, Tomohiro Hibino, Chunqi Zhao, Haoran Xie, Tsukasa Fukusato, and Kazunori Miyata. 2021. dualFace: Two-Stage Drawing Guidance for Freehand Portrait Sketching. *CoRR* abs/2104.12297 (2021). arXiv:2104.12297 <https://arxiv.org/abs/2104.12297>
- Ryoma Miyauchi, Tsukasa Fukusato, Haoran Xie, and Kazunori Miyata. 2021. Stroke Correspondence by Labeling Closed Areas. In *Nicograph International (NicoInt)*. IEEE, Tokyo, Japan, 34–41. <https://doi.org/10.1109/NICOINT52941.2021.00014>
- Wenwu Yang. 2017. Context-aware computer aided inbetweening. *IEEE Transactions on Visualization and Computer Graphics (TVCG)* 24, 2 (2017), 1049–1062. <https://doi.org/10.1109/TVCG.2017.2657511>
- Wenwu Yang, Hock-Soon Seah, Quan Chen, Hong-Ze Liew, and Daniel Šykora. 2018. FTP-SC: Fuzzy topology preserving stroke correspondence. *Computer Graphics Forum* 37, 8 (2018), 125–135. <https://doi.org/10.1111/cgf.13518>
- Lvmin Zhang, Chengze Li, Tien-Tsin Wong, Yi Ji, and Chunping Liu. 2018. Two-Stage Sketch Colorization. *ACM Trans. Graph.* 37, 6, Article 261 (Dec. 2018), 14 pages. <https://doi.org/10.1145/3272127.3275090>
- T. Y. Zhang and C. Y. Suen. 1984. A fast parallel algorithm for thinning digital patterns. *Commun. ACM* 27, 3 (1984), 236–239. <https://doi.org/10.1145/357994.358023>